ITA06-MACHINE LEARNING

M.BALAKUMARAN

192412179

Machine Learning Assessment Tool 2
Solved Answers (Any 5 Questions)

Q1. Machine Learning System Blocks

Machine Learning Pipeline:
Raw Data → Data Collection → Data Preprocessing → Feature Extraction → Model Selection → Model Training → Model Evaluation → Deployment → Prediction.

Supervised Learning:
• Uses labelled data.
• Used for classification and regression.
Example: Predicting Pass/Fail using student marks.

Unsupervised Learning:
• Uses unlabelled data.
• Finds hidden patterns or clusters.
Example: Customer segmentation using K-Means.

Difference:
Supervised: labelled data, predicts output.
Unsupervised: unlabelled data, discovers patterns.

Q3. Find-S Algorithm (Play Tennis)

Positive examples:
3:(Overcast, Hot, High, Weak)
4:(Rainy, Mild, High, Weak)
5:(Rainy, Cool, Normal, Weak)
7:(Overcast, Mild, Normal, Strong)

S0=<Ø,Ø,Ø,Ø>
After Ex3: <Overcast, Hot, High, Weak>
After Ex4: <?, ?, High, Weak>
After Ex5: <?, ?, ?, Weak>
After Ex7: <?, ?, ?, ?>

Final Hypothesis:<?, ?, ?, ?>

Q6. Find-S Algorithm (Workout)

Positive examples:
2:(Rainy, Mild, Evening, Happy)
3:(Sunny, Mild, Evening, Happy)
4:(Cloudy, Cool, Morning, Happy)

S0=<Ø,Ø,Ø,Ø>
After Ex2:<Rainy, Mild, Evening, Happy>
After Ex3:<?, Mild, Evening, Happy>
After Ex4:<?, ?, ?, Happy>

Final Hypothesis:<?, ?, ?, Happy>

Interpretation: Happy mood is the only common attribute.

Q11. Machine Learning Pipeline

Architecture:
Data Collection → Data Preprocessing → Feature Engineering →
Train/Test Split → Model Training → Model Evaluation → Deployment → Prediction.

Stages:
1. Data Collection
2. Data Preprocessing
3. Feature Engineering
4. Model Training
5. Model Evaluation
6. Deployment

Training Error:
Error on training data.

Generalization Error:
Error on unseen test data.

Example:
Training accuracy = 99%
Test accuracy = 92%
Difference indicates generalization performance.

Q15. Clustering in Unsupervised Learning

K-Means Steps:
1. Choose K.
2. Initialize centroids.
3. Assign points to nearest centroid.
4. Recompute centroids.
5. Repeat until convergence.

Example:
Data = {2,3,4,10,11,12}
K=2
Cluster1={2,3,4}
Cluster2={10,11,12}

Elbow Method:
Choose the value of K where WCSS begins to decrease slowly.

K-Means vs Hierarchical:
• K-Means requires K beforehand.
• Hierarchical does not.
• K-Means is faster.
• Hierarchical produces a dendrogram.